

YELP REVIEWS DATASET

NumPy and Pandas Problem Statements & Solutions

Data Analysis Assignment
20 Problem Statements with Clear Solutions

About This Assignment

This assignment contains 20 problem statements based on the Yelp Reviews dataset. Each problem is designed to help you learn NumPy and Pandas — two essential libraries for data analysis.

Dataset Overview

The Yelp Reviews dataset contains customer review information with the following columns:

- **review_id**: Unique identifier for each review
- **user_id**: Unique identifier for the reviewer
- **business_id**: Unique identifier for the reviewed business
- **stars**: Star rating given by the reviewer (1–5)
- **useful**: Number of 'useful' votes the review received
- **funny**: Number of 'funny' votes the review received
- **cool**: Number of 'cool' votes the review received
- **text**: Full text content of the review
- **date**: Date when the review was posted

These problems cover:

- Loading and viewing data with Pandas
 - Basic calculations (mean, median, min, max, std)
 - Filtering and sorting data
 - Grouping data and aggregations
 - Working with missing data
 - Date-time operations on review timestamps
 - Creating derived columns and boolean flags
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20 Problem Statements & Solutions

1. Load the Yelp Reviews dataset and display the first 5 rows of data.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
print(df.head())
```

2. Find the total number of reviews in the dataset.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
total_reviews = len(df)
print(f"Total reviews: {total_reviews}")
```

3. Calculate the average star rating across all reviews.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
avg_stars = df['stars'].mean()
print(f"Average star rating: {avg_stars:.2f}")
```

4. Find the review with the highest number of 'useful' votes.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
top_useful = df.loc[df['useful'].idxmax()]
print(f"Review ID: {top_useful['review_id']}")
print(f"Useful votes: {top_useful['useful']}")
```

5. Count how many unique businesses have been reviewed.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
unique_businesses = df['business_id'].nunique()
print(f"Unique businesses: {unique_businesses}")
```

6. Find the top 5 users who wrote the most reviews.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
top_users = df['user_id'].value_counts().head(5)
print(top_users)
```

7. Get the average star rating given by each user.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
user_avg_stars = df.groupby('user_id')['stars'].mean()
print(user_avg_stars)
```

8. Find all reviews with a 5-star rating.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
five_star = df[df['stars'] == 5]
print(f"5-star reviews: {len(five_star)}")
print(five_star[['review_id', 'stars', 'text']].head())
```

9. Find the minimum and maximum number of 'funny' votes on any review.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
min_funny = df['funny'].min()
max_funny = df['funny'].max()
print(f"Minimum funny votes: {min_funny}")
print(f"Maximum funny votes: {max_funny}")
```

10. Count how many reviews each star rating (1 to 5) has received.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
star_counts = df['stars'].value_counts().sort_index()
print(star_counts)
```

11. Calculate the standard deviation of the 'cool' votes column.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
std_cool = df['cool'].std()
print(f"Standard deviation of cool votes: {std_cool:.2f}")
```

12. Find all reviews posted between January 2020 and December 2022.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
df['date'] = pd.to_datetime(df['date'])
filtered = df[(df['date'].dt.year >= 2020) & (df['date'].dt.year <= 2022)]
print(f"Reviews between 2020-2022: {len(filtered)}")
```

13. Create a new column 'is_helpful' that is True if useful votes > 10.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
df['is_helpful'] = df['useful'] > 10
helpful_reviews = df[df['is_helpful'] == True]
print(f"Helpful reviews: {len(helpful_reviews)}")
```

14. Find the top 3 businesses that received the most reviews.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
top_businesses = df['business_id'].value_counts().head(3)
print(top_businesses)
```

15. Calculate the average number of 'useful' votes per review.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
avg_useful = df['useful'].mean()
print(f"Average useful votes per review: {avg_useful:.2f}")
```

16. Find and display reviews where the star rating is missing (NaN values).

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
missing_stars = df[df['stars'].isna()]
print(f"Reviews with missing stars: {len(missing_stars)}")
print(missing_stars[['review_id', 'stars']].head())
```

17. Sort reviews by the number of 'useful' votes in descending order and show the top 10.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
sorted_reviews = df.sort_values('useful', ascending=False)
print(sorted_reviews[['review_id', 'useful', 'stars']].head(10))
```

18. Find the median star rating of all reviews in the dataset.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
median_stars = df['stars'].median()
print(f"Median star rating: {median_stars:.2f}")
```

19. Count how many reviews have a star rating between 3 and 4 (inclusive).

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
mid_reviews = df[(df['stars'] >= 3) & (df['stars'] <= 4)]
print(f"Reviews with rating 3-4: {len(mid_reviews)}")
```

20. Get basic statistics (count, mean, std, min, max) for the 'stars' column.

Solution:

```
import pandas as pd
df = pd.read_csv('yelp_reviews.csv')
stats = df['stars'].describe()
print(stats)
```